



COEP TECHNOLOGICAL UNIVERSITY (COEP Tech)
A Unitary Public University of Government of Maharashtra
(Formerly College of Engineering Pune (COEP))

MID Semester Examination

Programme: B. Tech. S.Y.

Semester: Odd

Course Code:CT-20008

Course Name:PPL

Branch:Dept. of computer science and engineering

Academic Year: 2024-25

Duration: 1hr 30 min

Max Marks: 30

Student PRN No.

6	1	2	2	0	3	0	1	2
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Instructions:

1. Figures to the right indicate the full marks.
2. Mobile phones and programmable calculators are strictly prohibited.
3. Writing anything on question paper is not allowed.
4. Exchange/Sharing of stationery, calculator, etc. not allowed.
5. Write your PRN Number on the Question Paper.

			Marks	CO	PO
Q 1	a	In which architectural model are instructions and data stored in separate memory units? A) Von Neumann Architecture, which combines data and instructions B) Harvard Architecture, facilitating simultaneous access C) Modified Harvard Architecture, optimizing performance D) RISC Architecture, focusing on simplicity	1	1	1
	b	Which data structure ensures efficient access and manipulation capabilities within high-level programming environments? A) Arrays B) Linked Lists C) Trees D) All of the above	1	3	2
	c	In Compiler lexical analyser is used for? A) removing comments B) removing whitespace C) breaking the syntaxes in the set of tokens D) All of the mentioned	1	2	1
	d	Which programming languages are classified as low-level languages? A) COBOL, Fortran, Pascal B) prolog 2, expert systems C) assembly languages D) knowledge-based system	1	2	3

	e	Programming language FOTRAN stands for: A) free translator B) formula translator C) formula translation D) free translation	1	1	3
	f	Which of the following is not a quality of a programming language? A) Efficiency B) reliability C) readability D) Easy	1	2	1
	g	Keywords are recognized in a compiler during - A) the code generation B) the data flow analysis C) the lexical analysis of the program D) the program parsing	1	3	2
	h	The syntax is described by..... that defines the form of a language A) a set of rules B) keywords C) semantics D) Program	1	1	2
	i	A programmer writes a program to multiply two numbers instead of dividing them by mistake, how can this error be detected? A) Compiler or interpreter B) Compiler only C) Interpreter only D) None of the mentioned	1	1	1
	j	The output of the lexical analyser is _____ A) string character B) a syntax tree C) a set of RE D) a set of tokens	1	3	1
Q 2	a	What is a programming paradigm? Outline the differences between the imperative and declarative paradigms with examples.	2	2	3
	b	what are different implementation methods? State the difference between compiler and interpreter.	3	1	1
	c	What are compilers? State the phases involved in the compilation process. Include aspects in the lexical analysis phase.	5	3	2
Q 3	a	Discuss in detail the role of syntax and semantics in programming languages. Explain the language processors for C language code in short.	2	1	1
	b	Explain in short the different programming domains and programming languages used.	3	2	2
	c	Critically analyse the impact of various computer architectures on the design of programming languages. Explain the concepts in detail.	5	3	3